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> The factory system already has the relevant environment set up, so the installation steps can be skipped.

JetCam is an easy-to-use Python library developed by NVIDIA for the Jetson platform, which is used to integrate and operate USB cameras or CSI cameras

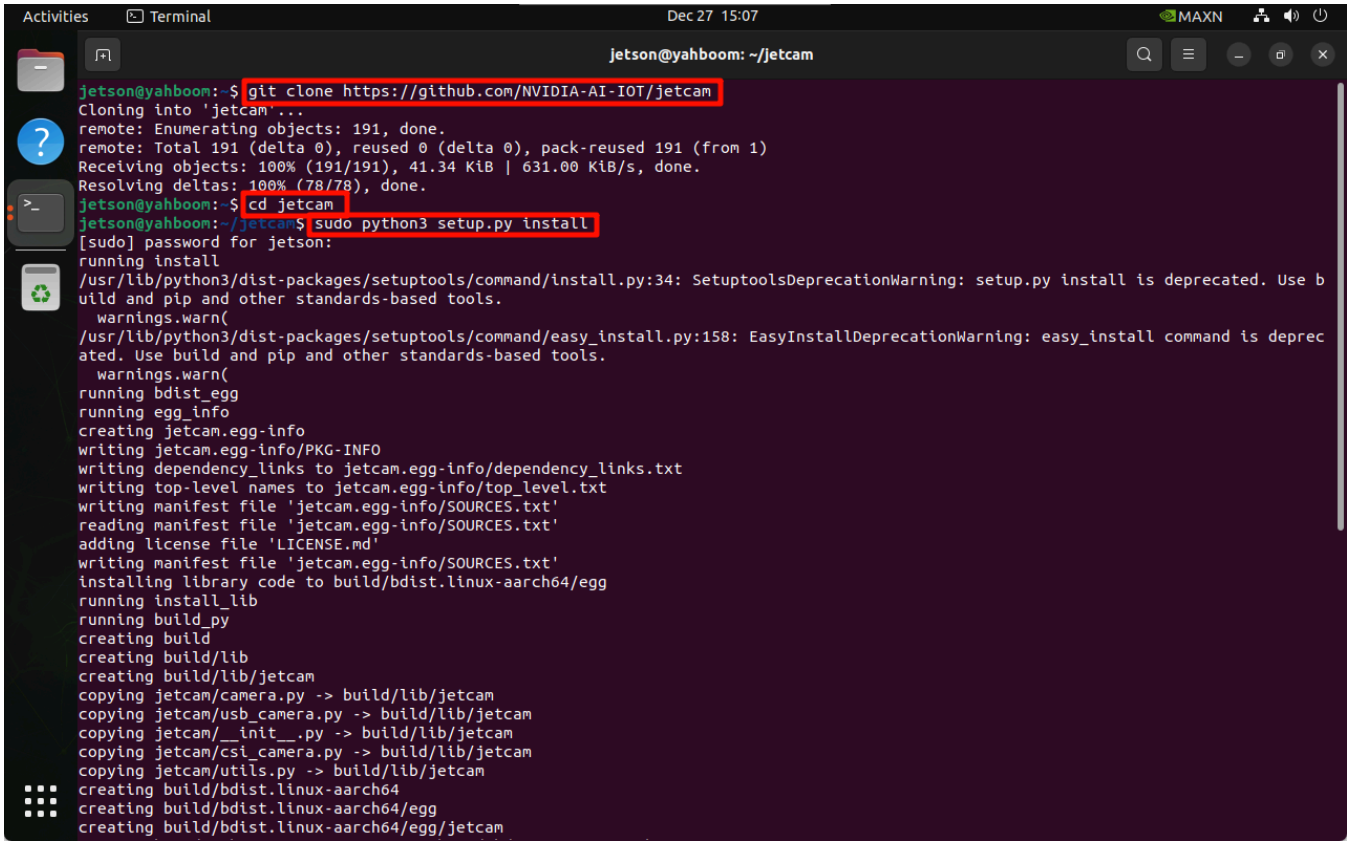
1. JetCam installation

```
git clone https://github.com/NVIDIA-AI-IOT/jetcam
```

```
cd jetcam
```

```
sudo python3 setup.py install
```

```
sudo pip3 install ipywidgets --break-system-packages
```



2. JetCam use

JetCam provides typical sample programs to demonstrate the calling of CSI and USB cameras to users.

```
The example needs to be run using Jupyter Lab. Using our factory image system, you can directly access it through the mainboard IP:8888!
```

2.1. CSI camera

Enter the folder where the CSI camera is located on the Jupyter Lab web page and open the corresponding folder:

```
/home/jetson/jetcam/notebooks/csi_camera
```

Note: If you are not familiar with Jupyter Lab, you can read the Jupyter Lab tutorial to learn the basic operations!

Main code explanation

Call the camera

width: image output width

height: image output height

```
from jetcam.csi_camera import CSICamera
camera = CSICamera(width=224, height=224)
```

Get the camera image

```
image = camera.read()
```

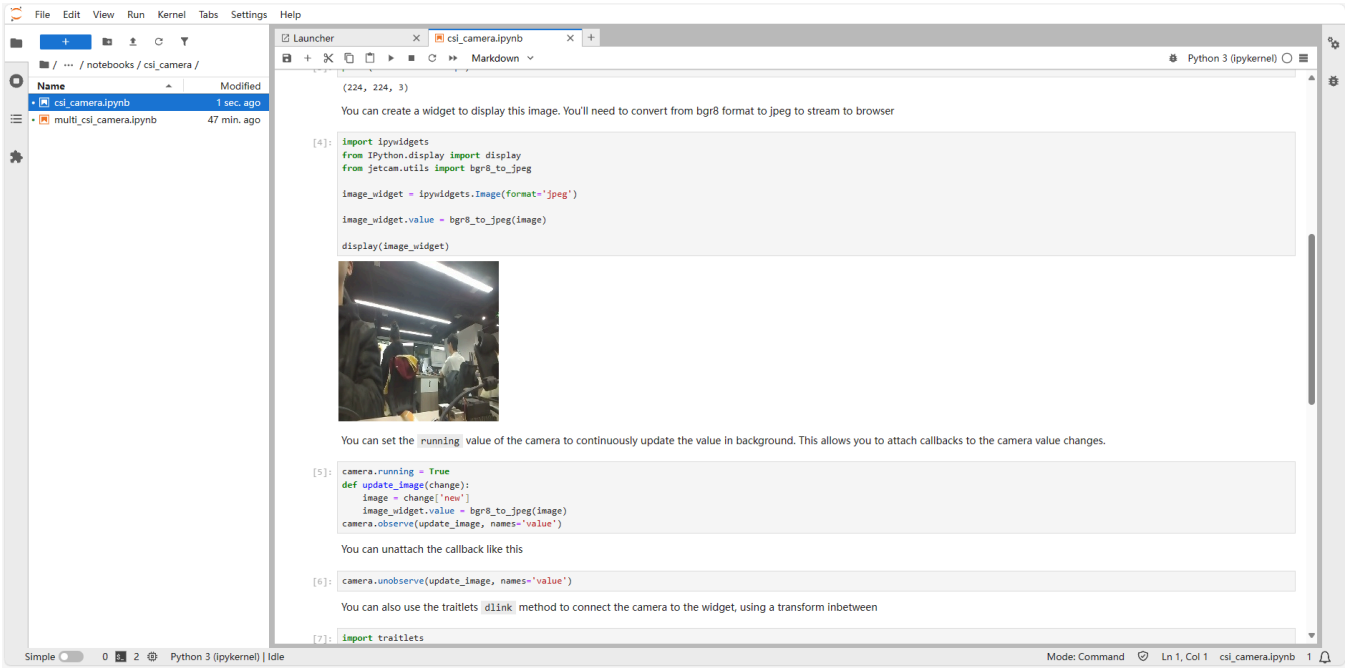
2.1.1. Single-channel camera

> **Source code path**

```
/home/jetson/jetcam/notebooks/csi_camera/csi_camera.ipynb
```

> **Running phenomenon**

After opening the program file, a single cell runs from top to bottom:



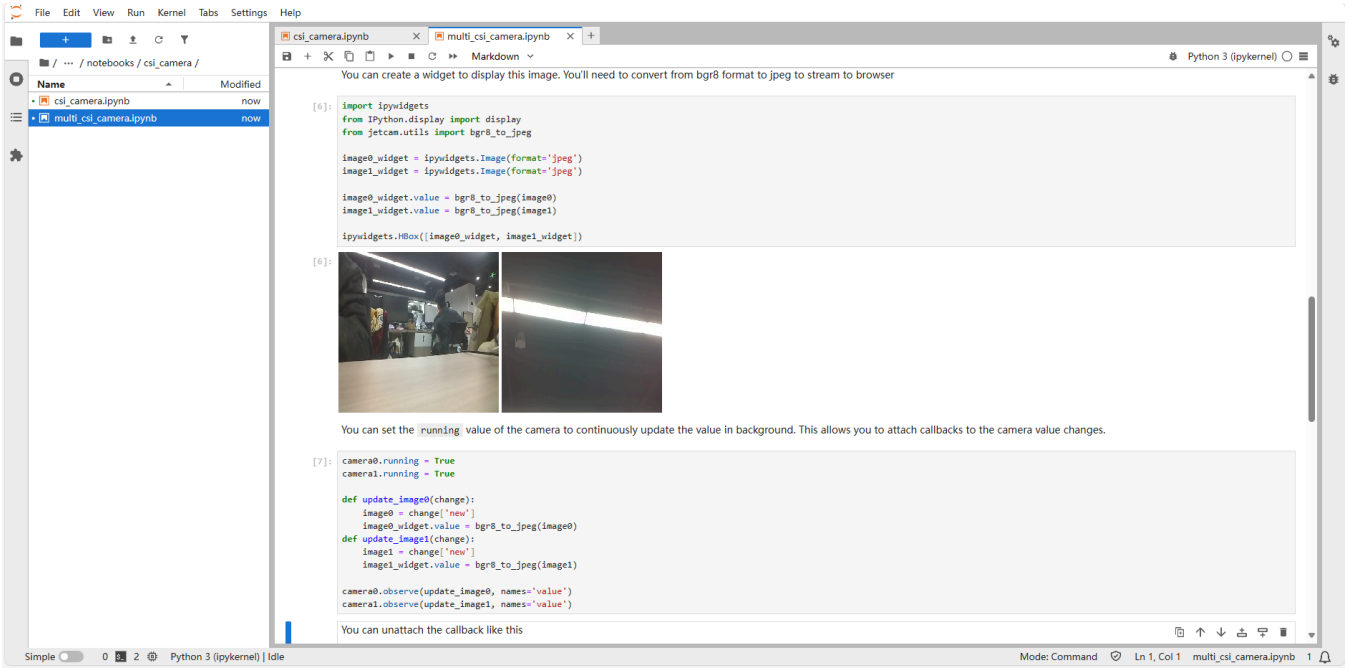
2.1.2. Multi-channel camera

> **Source code path**

```
/home/jetson/jetcam/notebooks/csi_camera/multi_csi_camera.ipynb
```

> **Running phenomenon**

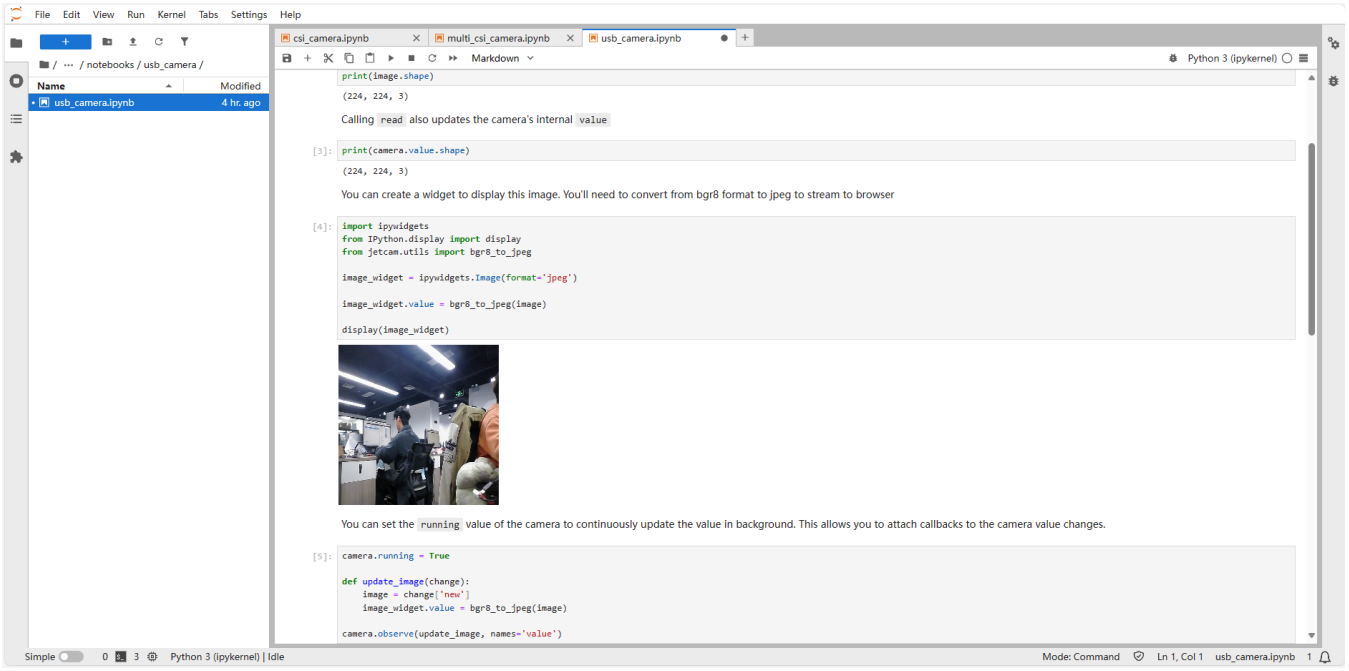
After opening the program file, a single cell runs from top to bottom:



2.2. USB camera

Jupyter Lab enters the folder where the USB camera is located and opens the file. The factory image system folder path is:

```
/home/jetson/jetcam/notebooks/usb_camera
```



References

https://github.com/NVIDIA-AI-IOT/jetcam